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Problem Solving Protocol

Multimodal geometric learning for antimicrobial peptide identification by leveraging alphafold2-predicted structures and surface features

Zehua Sun[1,2,‡], Jing Xu[2,3,‡], Yumeng Zhang[2,3], Yiwen Zhang[4], Zhikang Wang[2,3], Xiaoyu Wang[2,3], Shanshan Li[4], Yuming Guo[4], Hsin Hui Shen[1,2,*], Jiangning Song[2,3,*]

[1]Department of Materials Science and Engineering, Faculty of Engineering, Monash University, 20 Research Way, New Horizons, Clayton, Victoria 3800, Australia
[2]Biomedicine Discovery Institute and Department of Biochemistry and Molecular Biology, Monash University, Wellington Rd, Clayton, Melbourne, VIC 3800, Australia
[3]Monash Data Futures Institute, Monash University, Wellington Rd, Clayton, Melbourne, VIC 3800, Australia
[4]Climate, Air Quality Research Unit, School of Public Health and Preventive Medicine, Monash University, 553 St Kilda Road, Melbourne, VIC 3004, Australia

[*]Corresponding authors. Jiangning Song, Biomedicine Discovery Institute and Department of Biochemistry and Molecular Biology, Monash University, Victoria 3800, Australia. E-mail: Jiangning.Song@monash.edu; Hsin Hui Shen, Department of Materials Science and Engineering, Faculty of Engineering, Monash University, Clayton, Victoria 3800, Australia. E-mail: Hsin-Hui.Shen@monash.edu.
[‡]Zehua Sun and Jing Xu contributed equally to this work and should be considered co-first authors.

Abstract

Antimicrobial peptides (AMPs) are short peptides that play critical roles in diverse biological processes and exhibit functional activities against target organisms. While numerous methods have demonstrated the effectiveness of deep neural networks for AMP identification using sequence features; nevertheless, higher-level peptide characteristics—such as 3D structure and geometric surface features—have not been comprehensively explored. To address this gap, we introduce the SSFGM-Model (Sequence, Structure, Surface, Graph, and Geometric-based Model), a novel framework that integrates multiple feature types to enhance AMP identification. The model represents each peptide sequence as a graph, where nodes are characterized by amino acid features derived from ProteinBERT, ESM-2, and One-hot embeddings. Graph convolutional networks and an attention mechanism are employed to capture high-order structural and sequential relationships. Additionally, surface geometry and physicochemical properties are processed using a geometric neural network. Finally, a feature fusion strategy combines the outputs from these subnetworks to enable robust AMP identification. Extensive benchmarking experiments demonstrate that the SSFGM-Model outperforms current state-of-the-art methods. An ablation study further confirms the critical role of sequence, structural, and surface features in AMP identification. The key contribution of this work is the innovative integration of multiple levels of peptide characteristics and the combination of geometric and graph neural networks. This approach provides a more comprehensive understanding of the sequence-structure–function relationship of peptides, paving the way for more accurate AMP prediction. The SSFGM-Model has a significant potential for applications in the discovery and design of novel AMP-based therapeutics. The source code is publicly available at https://github.com/ggcameronnogg/SSFGM-Model.

Keywords: antimicrobial peptides; sequence features; surface features; structure information; geometric learning; AlphaFold2; graph neural network

Introduction

In recent years, the excessive and improper use of antibiotics has led to increasing antibiotic resistance of pathogenic bacteria. The development of antibiotic-resistance genes and multidrug resistance during bacterial evolution can greatly reduce the effectiveness of antibiotic treatments [1]. The escalation in the number of antibiotic-resistant bacterial isolates, accompanied by the bottlenecks in antibiotic discovery, has resulted in the resurgence of easy-treated diseases [2]. Consequently, antibiotic-resistant bacteria have emerged as a growing global health threat [3]. Meanwhile, antimicrobial peptides (AMPs) have emerged as a promising alternative to effectively combat antibiotic-resistant bacteria using unique antimicrobial mechanisms [4]. AMPs are a class of natural or synthetic peptide molecules. They are composed of short amino acid sequences, typically ranging from 10 to 50 amino acids. These peptides are widely distributed in nature. AMPs represent a crucial defense mechanism that organisms have developed through evolution [5]. Unlike traditional antibiotics that target specific sites, AMPs exhibit broad-spectrum activity by disrupting pathogen membranes or intracellular components, leading to cell death. Their versatility allows them to be integrated with advanced technologies, such as nanoparticle-based delivery systems, for clinical applications [6]. This significantly reduces the risk of emerging drug-resistant bacteria [7]. These highly diverse peptides, which leverage surface characteristics like polarity, disrupt the membranes of other organisms,

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ultimately inducing cell death [8]. Despite their high-antibacterial activity and functional diversity, the experimental discovery of AMPs is still hindered by efficiency and cost limitations. Therefore, an efficient computational platform that can accurately identify the antimicrobial properties of AMPs might accelerate the clinical application of AMP-based therapeutics [9].

The rapid growth of peptide datasets has spurred the creation of numerous peptide data platforms, such as the Peptide Utility search server, providing researchers with rich resources for AMP discovery [10]. Concurrently, deep learning models, combined with extensive datasets, have emerged as powerful computational tools in biological science, providing substantial savings in time and resources [11]. In recent years, AI-based approaches have been extensively applied to AMP identification [12]. For instance, tools like AMP Scanner [13], AMPfun [14], and DeepAmPEP30 [15] utilize a diverse machine learning and deep learning techniques, including classification and regression trees [16], convolutional neural networks (CNN) [17], and long short-term memory networks [18], to analyze AMP sequences and predict their antibacterial activities. These approaches primarily focus on manipulating sequence-based features for AMP classification and functional prediction.

However, AMPs are structurally diverse polymers, categorized by their secondary structures into α-helical, β-sheet, extended/random coil, and loop (cyclic) types [19]. Sequence-based representations are often insufficient for comprehensive analysis [20]. To address this limitation, graph neural networks (GNNs) appear as a robust approach for the structural modeling of AMPs [21]. GNN effectively incorporates spatial connections and adjacent information between atoms or amino acid nodes, enabling a more holistic representation of peptides. It also allows for a comprehensive understanding of how structural variations influence the peptide's antimicrobial activity [22]. For instance, sAMPpred-GAT [23] and esm-AxP-GDL [24] leveraged predicted peptide structures alongside GNN-based model to distinguish AMPs from non-AMPs. sAMPpred-GAT utilized trRosetta [25] for structure prediction, and then incorporated both structure and sequence information to enhance the attention-based learning for AMP classification [23]. Similarly, esm-AxP-GDL combined ESMFold-predicted structures with intermediate protein language model features extracted by ESM-2 [26] in the form of a graph-based model [24].

AMPs typically exert their antimicrobial effects by directly interacting with targets like bacterial membranes or proteins. Electrostatic attraction and hydrophobic interactions disrupt membranes or interfere with intracellular processes, ultimately causing bacterial cell death [12]. However, existing predictors did not consider the biophysical constraint and only modeled the amino acid level graph representing the whole AMP structure. We hypothesized that focusing on the functional surfaces of AMPs, particularly regions involved in protein-peptide interactions, could enhance predictions of AMP activity. Molecular Surface Interaction Fingerprinting (MaSIF) [20] provided a protocol to capture protein surfaces and compute a series of geometric and physiochemical features, including shape index, distance-dependent curvature, hydropathy, Poisson–Boltzmann continuum electrostatics, and free electrons/protons [20]. These features were extracted from AlphaFold2-predicted peptide structures, thereby supporting a more comprehensive analysis of AMP's functional properties.

In this study, we develop a multimodal geometric deep learning framework termed SSFGM-Model. It integrates the AlphaFold2-predicted structure with the pretrained ESM-2 and ProteinBERT representations into a GNN-based architecture, further enhanced by peptide surface fingerprints, to identify functional AMPs. We collect an extensive dataset of AMP sequences and structural data, providing a basis for representing the complex characteristics of these peptides. Furthermore, we perform extensive benchmarking experiments and ablation studies to illustrate the advantages of incorporating structural and surface inputs. As a result, SSFGM-Model demonstrates superior performance in predicting AMPs compared to the existing prediction tools. It may have practical implications for the discovery of novel AMPs and the development of future antimicrobial strategies.

Material and methods

Data collection and pre-processing

Benchmark dataset 1

The Benchmark Dataset 1 consists of 8222 peptide sequences from three subsets: (i) 2021 AMPs and 2021 non-AMPs retrieved from the AMP Scanner vr.2 web server (www.ampscanner.com), which included the updated sequences from the APD3 [27] database; (ii) 1100 unique AMP sequences derived from the original APD3 database [27]; (iii) 1536 AMPs and 1536 non-AMPs sourced from the reference (Xu *et al.*). The sequences in the third subset were aggregated from various resources, including ADAM [28], ADAPTABLE [29], APD3 [27], CAMP [30], dbAMP [31], DRAMP [32, 33], LAMP [34, 35], MilkAMP [36], and YADAMP [37]. We randomly shuffled the data from the three subsets and split the dataset into training (2878 AMPs and 2600 non-AMPs), validation (895 AMPs and 656 non-AMPs), and test sets (894 AMPs and 301 non-AMPs). The length distribution of the peptide sequences is illustrated in Fig. 1b (top). AMPs are generally shorter, primarily ranging from 25 to 50 amino acids, with a consistent distribution across the training, evaluation, and test sets. In contrast, non-AMPs exhibit a wider range of sequence lengths, spanning from 25 to nearly 150 amino acids. The test set was designed to assess the model's generalizability under the imbalanced data distribution.

Benchmark dataset 2

The Benchmark Dataset 2 was also derived from the AMP Scanner web server [13]. Unlike the updated version in Benchmark Dataset 1, it was frequently used for model training in recent deep-learning research on AMPs. Many computational tools have benchmarked their performance on this dataset. It consists of 1778 AMP and 1778 non-AMP sequences in which AMPs were primarily sourced from the APD3 database [27], targeting Gram-positive or Gram-negative bacteria. To ensure data quality, we confirmed that the dataset was curated by excluding sequences shorter than 10 amino acid residues and using the CD-HIT program to remove sequences with ≥90% sequence identity. We selected 2132 sequences (1066 AMPs and 1066 non-AMPs) for training and 1424 sequences (712 AMPs and 712 non-AMPs) for testing, as shown in Fig. 1a (right), which was consistent with the data splitting setting of the existing predictors. The non-AMPs were obtained from the UniProt [38] database excluding peptides with possible antimicrobial activities in the same manner as Benchmark Dataset 1. Additionally, non-AMPs shorter than 10 amino acid residues or sharing ≥40% sequence identity were removed. The length distribution patterns of AMPs and non-AMPs in Benchmark Dataset 2 were similar to those in Benchmark Dataset 1, as illustrated in Fig. 1b (bottom). It is important to note that while some peptide sequences overlap between Benchmark Dataset 1 and Benchmark Dataset 2, the training, validation, and test sets within each dataset are entirely independent. This ensures no overlap between the training and test sets, allowing for a fair and unbiased evaluation of the model's performance [13]. Benchmark dataset 2 was specifically used to compare our

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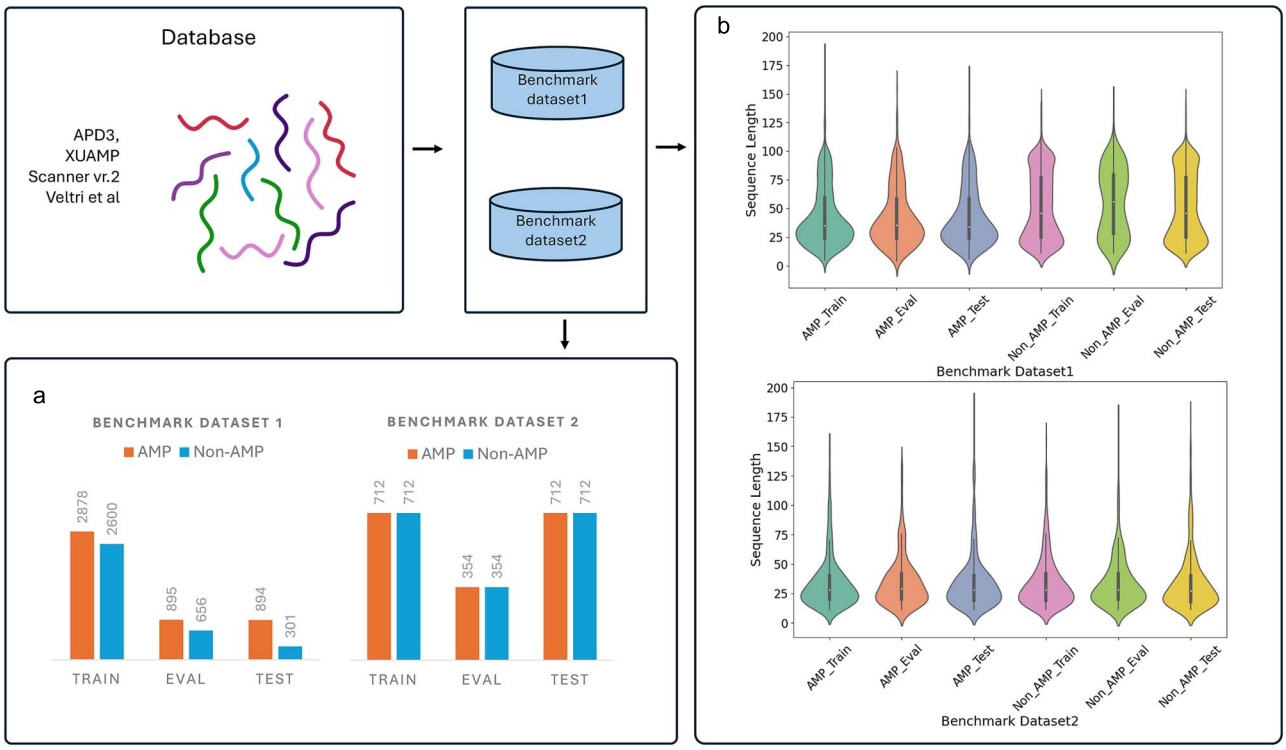

Figure 1. Overview of the benchmark datasets used in this study for AMP classification. (a) Summary of the number of samples in the training, validation, and test sets across the two benchmark datasets. (b) Distribution of sequence lengths for AMPs and non-AMPs in the training, validation, and test sets for both benchmark datasets. Benchmark Dataset 1 (top) exhibits a broader distribution of sequence lengths compared to Benchmark Dataset 2 (bottom).

model with existing AMP predictors that were trained and tested on the same dataset.

To further investigate the impact of structure prediction tools on AMP identification, we constructed Benchmark Dataset 3, which includes 7071 AMPs and 7071 non-AMPs derived from dbAMP database [39]. This dataset was designed to evaluate how the use of ESMFold, as an alternative to AlphaFold, influences AMP prediction performance. Detailed information about Benchmark Dataset 3 is provided in Supplementary Material Text S4.

The overview of SSFGM-model

The framework of SSFGM-Model is shown in Fig. 2. The input peptide sequences were embedded as pretrained feature vectors using two advanced protein language models, ProteinBert [40] and ESM-2 [26], which represent evolutionary information, along with One-hot encodings. AlphaFold2 [41] was employed to predict the 3D structures from peptide sequences. To construct the peptide graph, amino acid residues were selected as nodes and the residue-level embeddings were used as the corresponding node features. Structural relationships between amino acid residues were captured based on the pairwise Euclidean distance between Cα atoms. Specifically, when the distance between two residue nodes was below a specific threshold (e.g. 8 Å and 10 Å in this study), an edge was established between them. The distance between residues could also serve as the edge attribute. A GNN-based network consisting of six graph convolutional network (GCN) layers and one graph attention network (GAT) layer was further utilized to perform feature propagation on the peptide graph. On the other hand, we referred to MaSIF [20] to extract geometric features (shape index and distance-dependent curvature) and chemical features (hydropathy index, continuum electrostatics, and locations of free electrons and proton donors) [20] from the peptide surface patches. These features were mapped onto the geodesic polar coordinates to describe the relative positions within the local surface patch. The geometric neural network was subsequently applied to the learning the underlying biophysical representations. Finally, the model fused the outputs of the two subnetworks to enhance the prediction for antimicrobial activity.

In addition to the SSFGM-Model, we developed a sequence-feature-based model, SEQC-Model (Sequence-based CNN Model), which integrated sequence information using a two-layer CNN. Two structure-based models were also established in this study: (i) SSGT-Model (Sequence and Structure-based GCN-Transformer Model) combining three GCN layers, two transformer layers with eight heads, and two GAT layers; (ii) SSGA-Model (Sequence and Structure-based GCN with Attention Model) employing six GCN layers and a single GAT layer. These models were all evaluated to assess the contribution of the key components in the SSFGM model.

Features selection

Sequence-level embedding

One-hot encoding

One-hot encoding was employed in this study to represent input amino acid residues. Specifically, each amino acid residue is encoded as a binary vector of fixed length, where only one position corresponding to the specific residue is set to 1, while all other positions are set to 0. Consequently, an entire amino acid sequence can be encoded as a two-dimensional matrix of size $[L, N_{Onehot}]$ where L denotes the length of amino acid sequence and N_{Onehot} is 20, indicating the number of standard amino acids.

ProteinBERT

ProteinBERT is a large protein language model trained using self-supervised learning on 106 million protein sequences from the UniProtKB/UniRef90 database [40]. It is capable of handling long-range dependencies within protein sequences. The embedding

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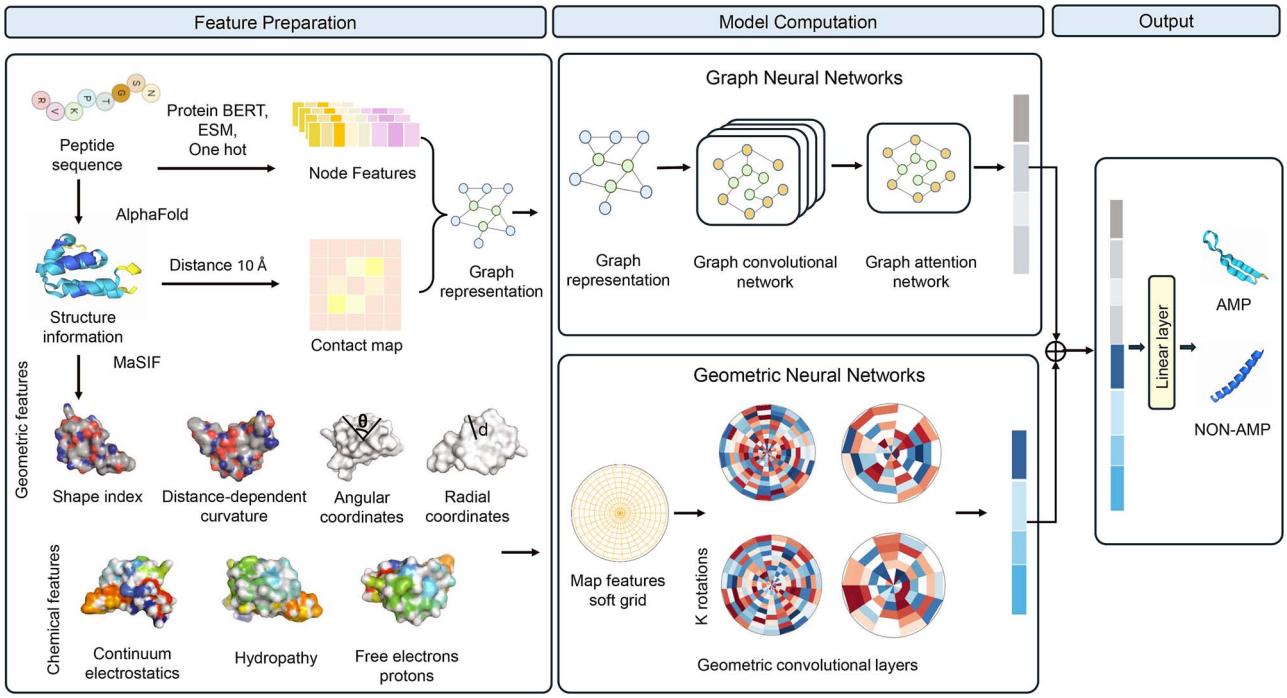

Figure 2. Schematic of the feature preparation, geometric learning, and model output for AMP prediction. (a) Feature generation pipeline. (b) Architecture of the graph neural network. (c) Surface features-based geometric neural network (d) Feature combination and model output.

vectors produced by ProteinBERT not only capture the contextual properties of the amino acid sequence but also encode each residue's physicochemical properties and potential interactions with distant residues [40]. We used the ProteinBERT pretrained embeddings in the form of a two-dimensional matrix of size [L, $N_{ProteinBERT}$] to represent peptides, where L denotes the length of amino acid sequence and $N_{ProteinBERT}$ is 1024.

ESM-2 embedding

The ESM-2 model utilizes self-supervised learning on large-scale protein sequence databases, primarily the UniRef [42] database and MGnify [43]. Protein sequences are processed through multiple layers of Transformer encoders, which employ self-attention mechanisms and feed-forward neural networks [44]. The model can generate latent feature representations for each amino acid residue [45], capturing both syntactic and semantic information within the sequence, as well as interactions between residues and three-dimensional structural information [26]. In this study, we employed ESM-2 3B model to embed the peptide sequences, representing each sequence as a two-dimensional matrix of size [L, N_{ESM2}] where L denotes the length of amino acid sequence and N_{ESM2} is 2560.

Structure-level embedding and graph construction

We employed the ColabFold [46] version of AlphaFold2 [41] to predict the three-dimensional structures of all peptide sequences in our dataset, generating corresponding Protein Data Bank (PDB) files containing atomic coordinates of the peptides, where a PDB file is a standard text format used to represent the three-dimensional structure of biological macromolecules [47]. Graphs were constructed based on these coordinates where the edges represent potential interactions between amino acid residues, such as hydrogen bonds, hydrophobic interactions, ionic bonds, and van der Waals forces [48]. The three-dimensional coordinates provide information about the distances between amino acid residues, reflecting their spatial interactions [49]. Thus, we extracted the three-dimensional coordinates of each peptide sequence from the predicted structure with the highest predicted Local Distance Difference Test (pLDDT) score [41]. The positions of the Cα atoms in all standard amino acids were selected to represent the residue-level coordinates. Next, edges were constructed in the graph based on the Euclidean distances between residues. By calculating the distances and applying a threshold (10 Å or 8 Å in this study), we determined whether an undirected edge should be established between two amino acid residues. The corresponding distance was also included as an edge attribute. This approach enabled the graph to reflect spatial proximity between residues, helping the GNN-based network capture structural interactions within the peptide.

With regard to the node features, we combined the sequence-level feature vectors, including One-hot encoding, ProteinBERT embeddings, and ESM-2 embeddings. Each node represented an amino acid residue, with its sequence features, while the edges captured the spatial interactions between residues. Using this approach, we generated peptide graphs that incorporate both sequence and structural information, which were used to train the graph neural network for AMP classification.

Surface-level features

The geometric and chemical features for each vertex on the protein surface were extracted from the three-dimensional structures of peptide sequences predicted by AlphaFold2. Each peptide covered several surface patches with a geodesic radius of $r = 12$Å and 100 vertices [20], And each vertex was characterized by five geometric or physiochemical properties: shape index, distance-dependent curvature, Poisson–Boltzmann continuum electrostatics, free electrons and proton donors, and hydropathy [20].

Shape index

The shape index represents the local geometric shape of each vertex, indicating the degree of concavity or convexity [50]. The

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closer the shape index value is to −1, the more concave the vertex is, while the closer it is to +1, the more convex the vertex is [20].

Distance-dependent curvature

Distance-dependent curvature represents the relationship between the distance from a vertex to the patch center and the surface normal at those points [50]. This relationship is expressed as a curvature value, reflecting the relative geometric shape between the vertex and the center. When the curvature value approaches −0.7, the region around the vertex is concave relative to the patch center; when the value approaches +0.7, the vertex is more convex compared to the patch center [20].

Poisson–Boltzmann continuum electrostatics

Poisson–Boltzmann continuum electrostatics is used to numerically represent the electrostatic properties of vertex on the meshed surface of the protein (Adaptive Poisson–Boltzmann Solver), Values closer to −1 indicate stronger negative charge at the vertex, while values closer to +1 indicate stronger positive charge [20].

Free electrons and proton donors

Free electrons and proton donors describe the potential of a protein surface vertex to act as a hydrogen bond donor or acceptor. Calculating the likelihood of each vertex being a free electron (hydrogen bond acceptor) or proton donor (hydrogen bond donor) helps the model identify regions on the protein surface that may participate in intermolecular interactions. Values closer to −1 indicate an optimal hydrogen bond acceptor site, while values closer to +1 indicate an optimal hydrogen bond donor site [20].

Hydropathy

The calculation of hydropathy is crucial for analyzing protein surface features. In this study, the hydropathy values are assigned to each vertex based on the Kyte and Doolittle scale of amino acid hydrophobicity [51]. The initial range of values is [−4.5, +4.5], which is then normalized to the range of [−1, +1]. In this normalized scale, values closer to −1 indicate a more hydrophilic vertex, while values closer to +1 indicate a more hydrophobic vertex [20].

Geodesic polar coordinates

The geodesic polar coordinates of each vertex included the radial coordinate and angular coordinate. The radial coordinate represented the distance from each vertex to the center of the patch, while the angular coordinate represented the direction of each vertex relative to the center. These vertex-based coordinates could capture not only the spatial features and geometric shape of each vertex but also their relative position to the patch center, laying the foundation for subsequent computations in the geometric neural network [20].

Model development

Graph neural networks

The GNN subnetwork in the SSFGM-Model consisted of three main components: GCN layers, GAT layers, and fully connected layers, as illustrated in Fig. 2b. We first concatenated three different embedding methods to create a 3604-dimensional feature for each node. Six consecutive GCN layers were used to propagate and update information from each node and its adjacent nodes through a weighted average aggregation. The impact of different GCN layer depths on model performance was systematically evaluated, as shown in Tables S6-1 and S6-2. They also progressively reduced the dimension of node features. It is implemented by combining the node's own features with a weighted average of its neighboring nodes' features.

$$H^{(l+1)} = \sigma\left(\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}H^{(l)}W^{(l)}\right),$$

where $H^{(l)}$ denotes the input feature matrix to the lth layer, $W^{(l)}$ denotes the weight matrix of the lth layer, $\sigma(\cdot)$ represents the ReLU activation function, $\tilde{A} = A + I$ is the weighted adjacency matrix with self-loop, and $\tilde{D}$ is the degree matrix corresponding to $\tilde{A}$.

Our model further incorporated a multi-head GAT layer to refine node feature representations. The GAT layer assigned different attention weights to each neighbor of a node, highlighting the important neighboring nodes.

$$h_i' = \|_{k=1}^{K}\sigma\left(\sum_{j\in\mathcal{N}(i)} e_{ij}\alpha_{ij}^{k}W^{k}h_j\right),$$

where h_i' denotes the output features of node i, $\mathcal{N}(i)$ denotes the set of neighbors of node i, e_{ij} is the edge weight between nodes i and j, W^k is the weight matrix of head k, α_{ij}^k is the attention coefficient for the edge connecting nodes i and j, and $\|$ represents the operation of concatenation.

$$\alpha_{ij}^{k} = \frac{\exp\left(\text{LeakyReLU}\left(a_k^T\left[W^k h_i \,\|\, W^k h_j\right]\right)\right)}{\sum_{j'\in\mathcal{N}(i)}\exp\left(\text{LeakyReLU}\left(a_k^T\left[W^k h_i \,\|\, W^k h_{j'}\right]\right)\right)},$$

where a_k^T represents the learnable attention vector of head k.

The GNN model architecture enabled effective learning of biological feature representation while reducing the data dimensionality. Node features were further aggregated into graph-level global features using global mean pooling, and then passed through fully connected layers to learn discriminative feature vectors.

Geometric neural networks

The geometric convolutional network in this study was based on the method described in the reference [20]. This model could interpret interaction fingerprints on protein molecular surfaces by integrating geometric and chemical features through geometric deep learning techniques. It identified surface structures by calculating shape index and distance-dependent curvature and detected interaction sites according to hydrophobicity index, electrostatic continuity, as well as the positions of free electrons and proton donors. Additionally, it utilized geodesic radius coordinates and central angular coordinates to pinpoint the positions of surface vertices.

The model employed Gaussian activation functions and rotation invariance techniques to perform multiple rotations [52]. After each rotation, it recalculated the spatial relationship between surface points and the central point and adjusted the angular coordinates accordingly. Gaussian functions were used to assign weights to each vertex, and features were aggregated across both rotations and vertices. Finally, these features were transformed through a fully connected layer. This rotation-invariant design ensured that key geometric and chemical features could be accurately captured, regardless of changes in the orientation of the protein surface.

Model training

To achieve our goal of recognizing AMPs, we employed a multimodal approach that integrated sequence, structure, and surface

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information through feature fusion (see the supplement file for more information). The fusion method combined output features from the GNN-based network and the geometric neural network. After feature extraction and concatenation, a ReLU activation function was employed to introduce nonlinear transformations, enhancing the model's ability to learn complex patterns. Subsequently, these processed features were fed to a linear layer and finally passed through a log softmax function to ensure the outputs could be interpreted as a probability distribution. A probability closer to 1 suggests the presence of an AMP, while a probability closer to 0 indicates a non-AMP.

For training the SSFGM-Model, the batch size was set to 32. The cross-entropy loss function [53] was employed for the classification task, while the AdamW optimizer [54] (with an initial learning rate of 0.0001 and a weight decay of 1e-2) was applied to update model parameters. Learning rate scheduling was managed using a combination of cosine annealing schedule [55] and reducing learning rate when a metric has stopped improving [56]. The learning rate was dynamically adjusted using a cosine annealing schedule over 10 epochs, with additional reductions triggered when the validation loss plateaued. Specifically, the learning rate was decreased by a factor of 0.1 after a patience period of 5 epochs. The model was trained for a total of 60 epochs, and a dropout rate of 0.5 was applied after each convolutional layer to mitigate overfitting. Training was performed on an NVIDIA GeForce RTX 3090 GPU, utilizing CUDA version 11.3, PyTorch version 1.12.1 [57], and Python version 3.10.12.

In this study, we evaluated three feature fusion techniques: direct concatenation of outputs from both models, linear transformation of the output features to a uniform dimension followed by concatenation, and postlinear transformation and concatenation using multihead attention mechanisms to selectively emphasize features more relevant to the task. The feature fusion techniques are described in detail in Supplementary Text S1–S3, and the results are shown in Tables S1 and S2. We assessed the performance under the best concatenation methods and have provided a detailed discussion in the "Result and Discussion" section of this paper.

Performance assessment

We employed multiple classification metrics to evaluate the performance of the model, including accuracy (ACC), sensitivity (SN), specificity (SP), precision (PR), Matthews' correlation coefficient (MCC), area under the ROC curve (AUC), and area under the precision-recall curve. ACC assesses the overall performance of the model in distinguishing AMPs (positive class) from non-AMPs (negative class). AUC provides a more comprehensive evaluation of the model's ability to separate positive and negative samples. The ROC curve illustrates the trade-off between sensitivity and specificity at various classification thresholds, where a higher AUC indicates stronger discriminatory power between antimicrobial and non-AMPs.

$$ACC = \frac{TP + TN}{TP + TN + FP + FN}$$

$$SENS = \frac{TP}{TP + FN}$$

$$SPEC = \frac{TN}{TN + FP}$$

$$PR = \frac{TP}{TP + FP}$$

$$MCC = \frac{(TP \times TN) - (FP \times FN)}{\sqrt{(TP + FP)\,(TP + FN)\,(TN + FP)\,(TN + FN)}}$$

The metrics are derived from ***TP*** (true positives), ***TN*** (true negatives), ***FP*** (false positives), and ***FN*** (false negatives), which collectively contribute to the evaluation. In cases of class imbalance, ***MCC*** offers a more reliable performance assessment by considering all four components (TP, TN, FP, FN), making it a better metric than ACC in such scenarios.

Result and discussion

Impact of the distance thresholds, and GCN and transformer layers for predictive performance

Firstly, we conducted a comparative analysis to evaluate the impact of varying distance thresholds, the number of GCN layers [58], and transformer layers [59] on the predictive performance of our models. Benchmark Dataset 1 was used for this comparison, and the results are summarized in Table 1. Specifically, we examined distance thresholds of 8 Å and 10 Å, as these values are often utilized to capture meaningful structural proximity in molecular interactions [60]. Two deep learning architectures (SSGT-Model and SSGA-Model) were compared in this study. For a fair and consistent comparison, both models were trained for 30 epochs, ensuring an equitable assessment of their predictive capabilities.

Figure 3a presents the performance metrics—ACC, AUC, MCC, SENS, and SPEC—obtained by the models using two network architectures with distance thresholds set at 8 Å and 10 Å. Notably, when employing an architecture that combines three GCN layers, two transformer layers with eight heads, and two GAT layers (SSGT-Model), the model with a 10 Å distance threshold exhibited superior performance, achieving an ACC of 0.8871, an AUC of 0.8897, and an MCC of 0.7731. These metrics surpassed those of the corresponding model with an 8 Å distance threshold, which achieved an ACC of 0.8774, an AUC of 0.8807, and an MCC of 0.7545. A similar trend was observed with architecture incorporating six GCN layers and a single GAT layer. The model using a 10 Å distance threshold outperformed the 8 Å model, achieving an ACC of 0.8929, an AUC of 0.9547, and an MCC of 0.7854, compared to the 8 Å model's ACC of 0.8871, AUC of 0.8897, and MCC of 0.7731. In particular, the AUC values for the 10 Å models substantially exceeded those of the 8 Å models. This marked improvement in predictive performance at the 10 Å distance threshold highlighted its ability to capture molecular interactions. Therefore, we selected the 10 Å distance threshold for subsequent model comparisons and evaluations, as it clearly provides a more effective representation for improving prediction outcomes.

Additionally, we compared the performance of the two model architectures (SSGA-Model and SSGT-Model). The results demonstrate that the architecture with six GCN layers and a single attention layer significantly outperformed the three-GCN-layer Transformer-based model. As shown in Table 1, at a distance threshold of 8 Å, the six-GCN-layer model achieved a higher AUC of 0.8897 and an ACC of 0.8774, compared to the Transformer-based model's AUC of 0.8807 and ACC of 0.8871. When the distance threshold was increased to 10 Å, the six-GCN-layer model further excelled, achieving an AUC of 0.9547 and an ACC of 0.8929, clearly surpassing the Transformer-based model, which had an AUC of 0.9137 and an ACC of 0.8875. These findings highlighted the advantage of using deeper GCN layers in combination with a GAT layer to boost predictive performance for this task. Consequently, we selected the six-GCN-layer architecture with a single GAT layer as the final module in further evaluations.

In summary, both the distance threshold and network architecture played critical roles in determining predictive accuracy. The 10 Å distance threshold consistently yielded superior results

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Table 1. Performance comparison of different structure-based models under varying thresholds.

Model	Distance	ACC	AUC	SENS	SPEC	MCC
SSGT-Model	8 Å	0.8774	0.8807	0.8597	0.9024	0.7545
SSGA-Model	8 Å	0.8871	0.8897	0.8725	0.9070	0.7731
SSGT-Model	10 Å	0.8875	0.9137	0.8658	0.9162	0.7745
SSGA-Model	10 Å	**0.8929**	**0.9547**	**0.8747**	**0.9177**	**0.7854**

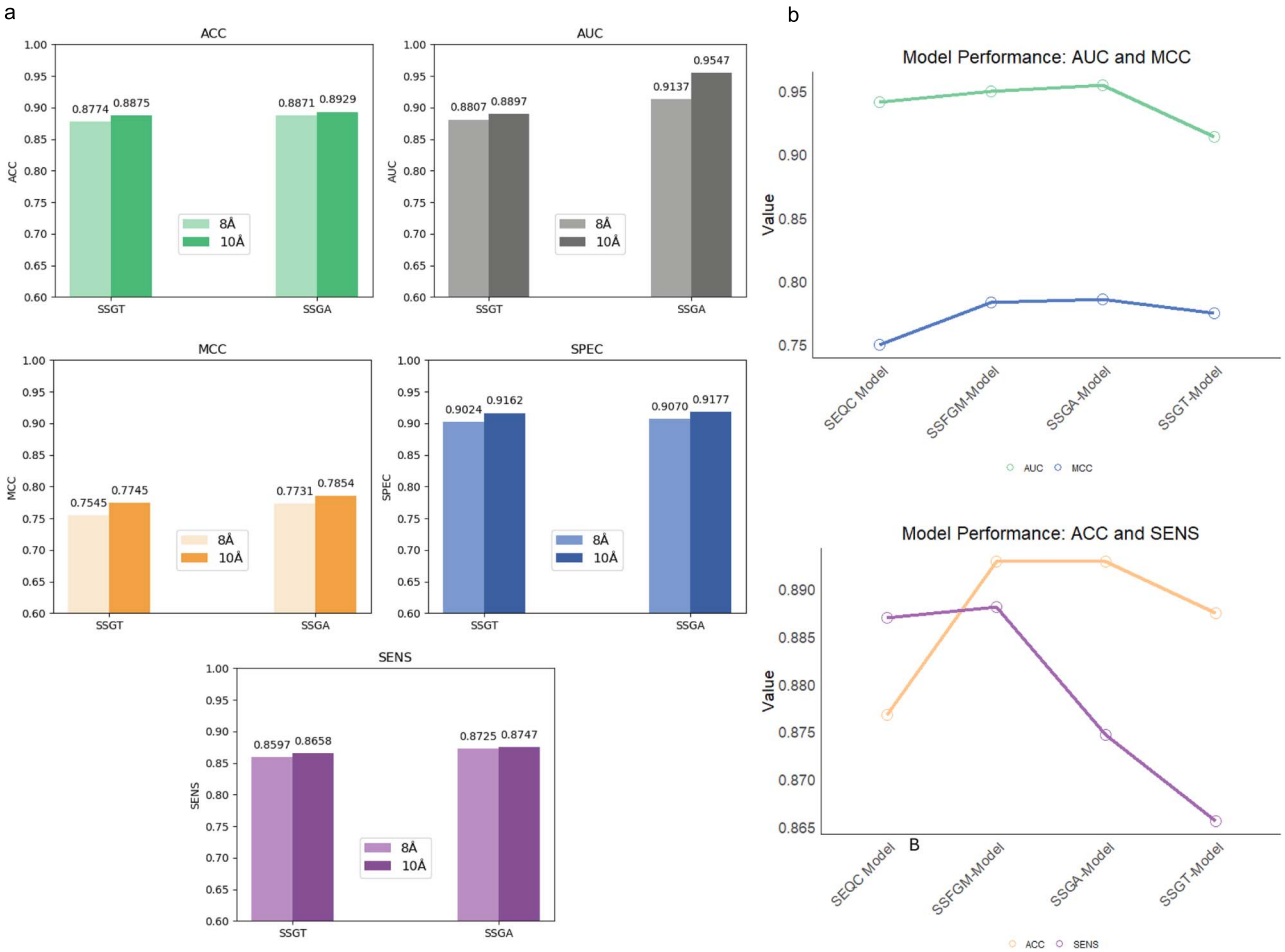

Figure 3. Performance comparison of structure-based models under different thresholds. (a) Bar plots displaying the performance of the SSGT and SSGA models under the thresholds of 8 Å and 10 Å, evaluated using five metrics: Accuracy (ACC), area under the ROC curve (AUC), Matthews correlation coefficient (MCC), specificity (SP), and sensitivity (SN). (b) Line graphs comparing the performance of multiple models, including the SEGC-model, SSGA-model, SSGT-model, and SSFGM-model.

across different models, and the architecture with six GCN layers and a GAT layer outperformed the more complex Transformer-based model.

Impact of peptide features on predictive performance

As mentioned above, we employed five types of peptide representation features as inputs to our model, including One-hot encodings, ProteinBERT encodings, ESM encodings, structure information of peptides (contact maps), and protein surface features derived from the AlphaFold2-predicted 3D structures. To investigate the combined contributions of these features to predictive performance, we first conducted a series of comparisons using various feature combinations based on Benchmark Dataset 1. Each model was trained for 50 epochs, and the results are summarized in Table 2 and Fig. 3b. Our results demonstrated that utilizing ProteinBERT, One-hot, and ESM-2 encodings through a network with two CNN layers (SEQC) achieved an AUC of 0.9415 and an ACC of 0.8768. Additionally, incorporating structural contact maps to construct peptide graphs, along with ProteinBERT, One-hot, and ESM encodings as node features through GCN layers (SSGA), further improved the model performance. The architecture with six GCN layers and a single GAT layer resulted in an AUC of 0.9547, an ACC of 0.8929, and an MCC of 0.7854. Meanwhile, an architecture (SSGT) comprising three GCN layers, two transformer layers with eight heads, and two GAT layers yielded an ACC of 0.8875 and an MCC of 0.7745, although the AUC was slightly lower (0.9137). These findings underscore the importance of incorporating both sequence and structural information to improve the model capacity.

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Table 2. Performance comparison of different models under the 10 Å threshold on benchmark dataset 1.

Model	ACC	AUC	SENS	SPEC	MCC
SEQC-Model	0.8768	0.9415	0.8870	0.8765	0.7496
SSGA-Model	0.8929	0.9547	0.8747	0.9177	0.7854
SSGT-Model	0.8875	0.9137	0.8656	0.9162	0.7745
SSFGM-Model	0.8929	0.9498	0.8881	0.8994	0.7829

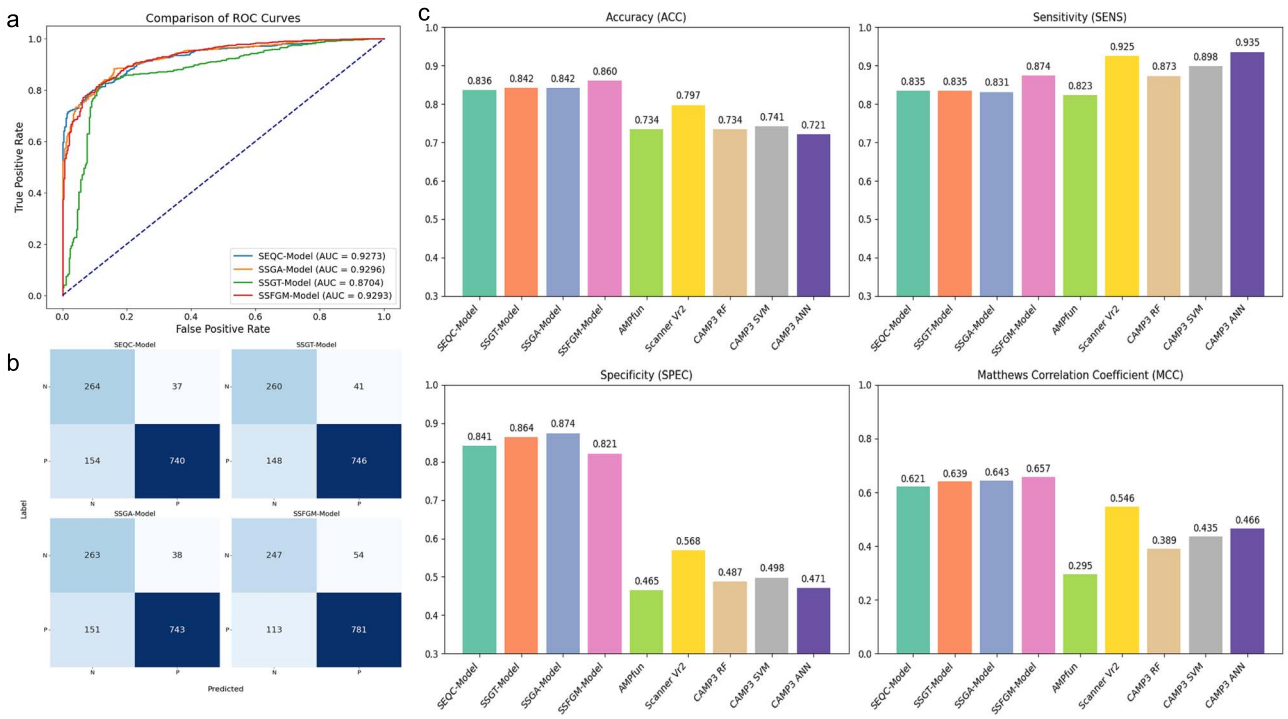

Figure 4. Performance comparison of AMP prediction models. (a) ROC curves for four models: SEQG-model, SSFGM-model, SSGA-model, and SSGT-model. (b) Confusion matrices for the four models, where "P" and "N" represent positive and negative classes. Each matrix displays the correct and incorrect classifications for each model, highlighting the improved balance of correctly classified samples in the SSFGM-model and SSGA-model. (c) Bar plots summarizing the performance of each model across four metrics: Aaccuracy (ACC), sensitivity (SN), specificity (SP), and Matthews' correlation coefficient (MCC). The performance of the proposed models is compared against existing web servers (AMPFun, scanner Vr2, CAMP3 RF, CAMP3 SVM, CAMP3 ANN).

When we incorporated protein surface features into the model (SSFGM), it reached the highest ACC of 0.8929, matching the performance of the architecture with six GCN layers and a single attention layer. However, despite this, the model's AUC and MCC values did not surpass those of the six-GCN-layer architecture. Specifically, the AUC only reached 0.9498, and the MCC stood at 0.7829, both slightly lower than the AUC of 0.9547 and MCC of 0.7854 achieved by the six-GCN-layer model without the surface features. Additionally, we evaluated these models on the benchmark dataset 1 test dataset. The ROC curves are presented in Fig. 4a, while the confusion matrices of different models are shown in Fig. 4b. The metrics are summarized in Fig. 4c. The model (SEQC) that used ProteinBERT, One-hot, and ESM encodings achieved an ACC of 0.836, an AUC of 0.9273, and an MCC of 0.6205. When incorporating the structural contact maps using the six-GCN-layer architecture, the model (SSGT) improved slightly, achieving an ACC of 0.8418, an AUC of 0.9296, and an MCC of 0.6430. Furthermore, when protein surface features were added to the model (SSFGM), a noticeable enhancement in predictive performance was observed. The ACC increased to 0.8603, the AUC reached 0.9293, and the MCC improved to 0.6568. Incorporating protein surface features improved the model's predictive accuracy and confidence, as indicated by the increased MCC.

Despite marginal improvements in AUC values, the overall discriminatory power of the models remained strong even without the surface features. However, the noticeable boost in MCC and ACC highlighted that surface features could capture additional molecular structural information, potentially enhancing classification performance. We also compared our model with several state-of-the-art (SOTA) models, including AMPfun [14], AMPScanner Vr2 [13], CAMP3-RF, CAMP3-SVM, and CAMP3-ANN [61], to evaluate predictive performance. AMPfun achieved an ACC of 0.797, an MCC of 0.546, and an SP of 0.821, performing moderately across multiple evaluation metrics. Although CAMP3 ANN showed the best SN at 0.935, effectively identifying positive samples, its performance in terms of ACC, SP, and MCC was relatively weaker, with values of 0.721, 0.471, and 0.466, respectively. In addition, while Scanner V2 exhibited high SN (0.925), its SP was only 0.465, and it had the lowest MCC of 0.295, indicating significant deficiencies in distinguishing negative samples and in overall classification ability. Similarly, CAMP3 RF and CAMP3 SVM also showed lower performance, with MCC values of 0.546 and

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Table 3. Performance comparison of different models on benchmark dataset 2.

Model	ACC	AUC	SENS	SPEC	MCC
SEQC-Model	92.49	0.9631	0.8778	0.9719	0.8535
SSGT-Model	93.61	0.9614	**0.9256**	0.9466	0.8724
SSGA-Model	93.82	0.9724	0.9017	0.9747	0.8788
SSFGM-Model	**93.89**	**0.9732**	0.9003	**0.9775**	**0.8804**

0.435, respectively. SSFGM-Model outperformed the others across multiple evaluation metrics, particularly excelling in ACC, SP, and MCC, demonstrating its superior capability to distinguish between positive and negative samples with higher specificity, reducing false positives. While CAMP3 ANN excelled in sensitivity, it lagged behind in terms of other performance metrics, and Scanner V2, despite its high sensitivity, exhibited significantly lower specificity and MCC. In conclusion, integrating protein surface features into the model provided a modest but meaningful improvement in predictive accuracy and confidence, highlighting the value of leveraging diverse structural information. These results highlight the importance of multimodal representations, especially those derived from 3D structures, for peptide prediction tasks. The combination of features from different modalities outperforms individual representations, providing a more comprehensive understanding of peptide properties.

To further evaluate the impact of structural predictions on model performance, we conducted a comparative analysis using structures predicted by AlphaFold2 and ESMFold on the validation and test sets of Benchmark Dataset 1. Unlike AlphaFold2, which relies on multiple sequence alignments (MSAs), ESMFold is a fast protein structure prediction model that does not require MSAs, making it computationally more efficient. The complete comparison results, including various evaluation metrics, are detailed in Supplementary Text S7.

The results revealed that while ESMFold-predicted structures enhanced the model's classification ability, they were slightly inferior to those predicted by AlphaFold2. For instance, in the SSGT-Model, ESMFold-predicted structures achieved a higher SP of 0.9375, whereas AlphaFold2-predicted structures resulted in higher SN (0.8656 vs 0.8568). This suggests that the ESMFold-based model was more effective in reducing false positives, while the AlphaFold2-based model excelled in identifying AMPs. For the SSGA-Model, the AlphaFold2 version outperformed the ESMFold version in terms of AUC (0.9547 vs 0.9514), SN (0.8747 vs 0.8557), and ACC (0.8929 vs 0.8903), indicating that AlphaFold2-derived structures provided superior overall classification performance. Similarly, in the SSFGM-Model, which integrates structural and surface features, the AlphaFold2 version demonstrated better SN (0.8881 vs 8680) and MCC (0.7829 vs 0.7783), while the ESMFold version achieved higher SP (0.9177 vs 0.8994) (Table S3-1 and Figure S3-1). These findings suggest that AlphaFold2-predicted structures enhance AMP detection, whereas ESMFold-predicted structures are more effective in minimizing false positives.

Despite the slight performance gap, the SSFGM-Model incorporating ESMFold structures still outperformed all sequence-only models on the test set of Benchmark Dataset 1, achieving the highest ACC (0.8410) and MCC (0.6404), along with superior SN (0.9311). This performance surpassed that of AMPScanner Vr2 (SN: 0.925) and the SEQC-Model (SN: 0.835) (Table S3-2 and Figure S3-2).

In conclusion, while AlphaFold2-predicted structures generally deliver better performance, ESMFold serves as a computationally efficient alternative in scenarios where AlphaFold2 is unavailable or computational resources are limited. ESMFold ensures stable model performance, making it a viable substitute for AlphaFold2. Complete data are available in Supplementary Material Text S7.

Comparative analysis based on training and Benchmark Dataset 2

For a fair comparison, we used the training and benchmark dataset constructed by Veltri *et al.* [13], which is widely recognized and adopted for training and evaluating various SOTA models. Additionally, we evaluated the effectiveness of different feature combinations on this benchmark dataset. The ACCs, AUCs, and MCCs are presented in Table 3 with the corresponding ROCs and precision-recall curves shown in Fig. 5a. Notably, the model that incorporated protein surface features into the GCN architecture achieved the highest performance, with an accuracy of 0.9389, an AUC of 0.9732, and an MCC of 0.8804. These results show a slight improvement over the version of the model without surface features, which had an accuracy of 0.9382, an AUC of 0.9724, and an MCC of 0.8788. Both models demonstrated superior performance compared to others. A more detailed comparison of model performance using different node embeddings, including one-hot, Protein-BERT, ESM-embedding, and their fusion, is provided in Table S7.

This enhanced performance highlighted the significant contribution of protein surface features in refining the model's predictive capabilities. The increases in AUC and MCC suggest that protein surface features provide the model with additional structural insights. This allows for a better capture of nuanced molecular interactions. This result underscored the importance of incorporating structural information beyond sequence-based features to achieve optimal performance.

Furthermore, we benchmarked our framework against several other SOTA models, as shown in Fig. 5b and Table 4. Both variants of our model—one with protein surface features and one without—outperformed the competing models in terms of ACC, AUC, and MCC. In this comprehensive evaluation of AMP prediction models, we observed the performance of traditional machine learning-based CAMP3 [30] series models, including CAMP3-ANN [30], CAMP3-Discriminant Analysis (DA) [30], CAMP3-Random Forest (RF) [30], and CAMP3-SVM [30]. These models exhibited certain limitations, with CAMP3-ANN achieving an ACC of 0.8405 and an MCC of 0.6813, CAMP3-DA recording an ACC of 0.8391 and an MCC of 0.6797, CAMP3-RF showing an ACC of 0.8757 and an MCC of 0.7553, and CAMP3-SVM posting an ACC of 0.8455 and an MCC of 0.6933. This reflects that traditional machine learning methods still have limitations in accurately predicting complex biological data. In contrast, deep learning-based models such as AMPScan Vr2, AMPScanner, iAMPCN [62], and ACEP [63] demonstrated superior performance. AMPScan Vr2 and AMPScanner both showed good ACC and SP, achieving 0.9101, 0.9213, and 0.9129, 0.9269, respectively, but their MCC values of 0.8204 and 0.8261 suggest there is potential for improvement in overall

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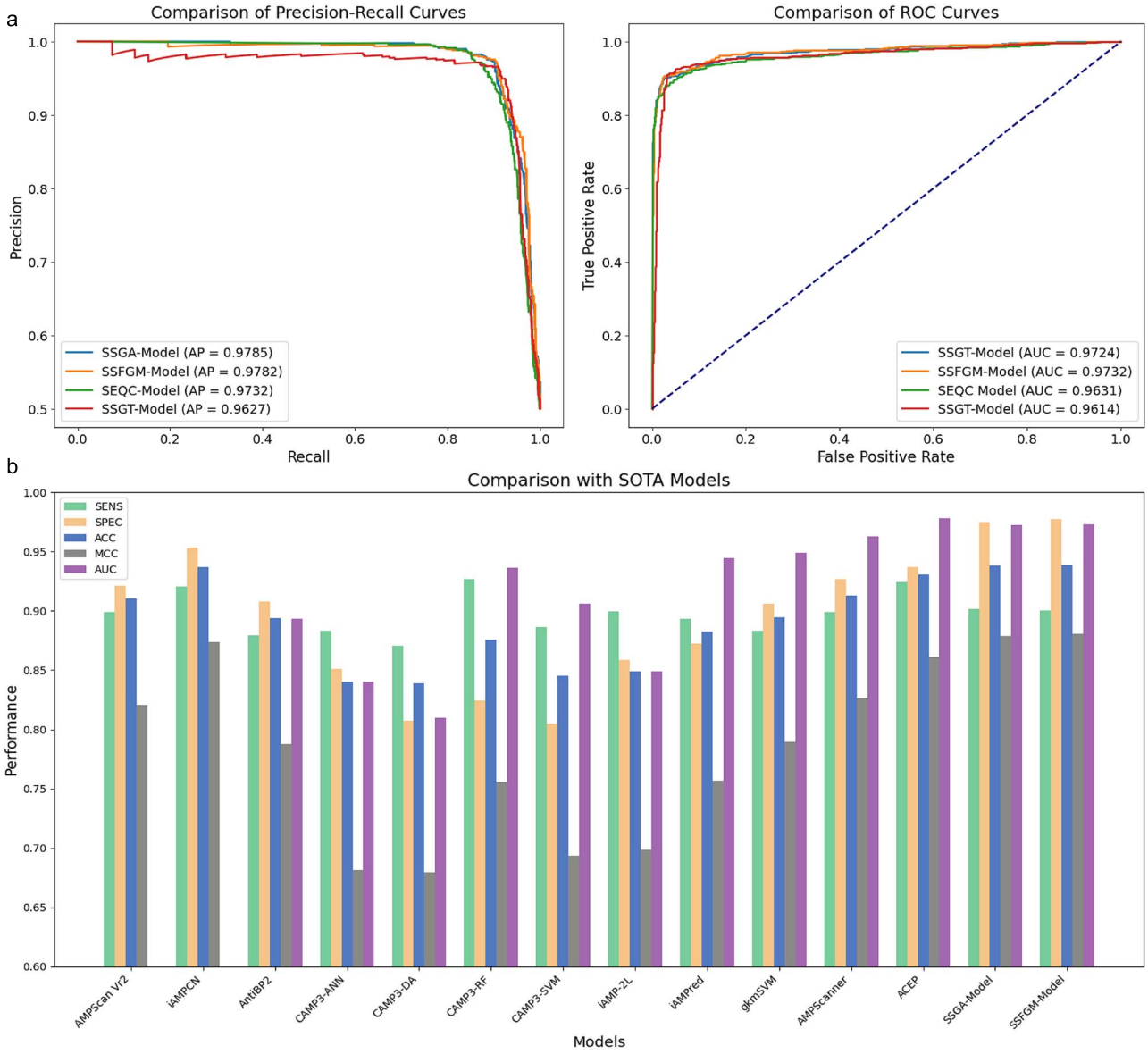

Figure 5. Performance comparison of different models on Benchmark Dataset 2. (a) Precision-recall and ROC curves for the four models (SSGT, SSFGM, SEQC, SSGA) evaluated on Benchmark Dataset 2. (b) Bar chart comparing the performance of the four proposed models against other SOTA models for AMP prediction on Benchmark Dataset 2.

prediction quality. Particularly, iAMPCN, with an ACC of 0.9368 and SP of 0.9531, approached the performance of top models, yet its MCC of 0.874 indicates a slight lack in predictive balance. The ACEP model, utilizing CNNs and attention mechanisms, exhibited an exceptionally high AUC of 0.9778 and a solid MCC of 0.861, proving its excellent capability in highly complex prediction tasks. Among these models, SSGA-Model and SSFGM-Model stand out, not only for their ACC and SP, which are 0.9382, 0.9747, and 0.9389, 0.9775, respectively, but also for their high MCC values of 0.8788 and 0.8804. Both models employ GCN combined with attention mechanisms, and the SSFGM-Model further integrates surface features and geometric neural networks. This advanced integration gives the SSFGM-Model a significant advantage in accurately identifying non-AMPs, effectively reducing false positive rates. This is crucial for the selection and research of AMPs. Moreover, their high MCC values not only demonstrate their excellent performance on individual metrics but also show outstanding balance in overall predictive performance. By understanding the spatial structure and surface chemical properties of molecules, the SSFGM-Model significantly enhances the precise prediction of AMP activity. The inclusion of protein surface features provided a modest but meaningful improvement, further enhancing the model's predictive power.

These results underscored the robustness of our approach. Even without protein surface features, the model performed exceptionally well, but incorporating these features yielded additional performance gains, making the model even more effective. This demonstrated the model's inherent strength and the potential for enhancing its predictive capabilities through adding structural information like protein surface features.

The analysis of surface feature importance revealed that Free Electrons and Proton Donors contributed the most significantly to model performance, followed by Poisson–Boltzmann continuum electrostatics and Hydropathy (Fig. S2). In contrast, shape index and distance-dependent curvature, while having

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Table 4. Performance comparison of our model and other SOTA models on Benchmark Dataset 2.

Model	ACC	AUC	SENS	SPEC	MCC
AMPScan Vr2	0.9101	—	0.8989	0.9213	0.8204
iAMPCN	0.9368	—	0.9203	0.9531	0.874
AntiBP2	0.8937	0.8936	0.8791	0.908	0.7876
CAMP3-ANN	0.8405	0.8405	0.883	0.8511	0.6813
CAMP3-DA	0.8391	0.8097	0.8707	0.8075	0.6797
CAMP3-RF	0.8757	0.9363	**0.9269**	0.8244	0.7553
CAMP3-SVM	0.8455	0.9062	0.8862	0.8047	0.6933
iAMP-2 L	0.849	0.849	0.8999	0.8586	0.6983
iAMPred	0.8827	0.9444	0.8933	0.8722	0.7566
gkmSVM	0.8946	0.949	0.8834	0.9059	0.7895
AMPScanner	0.9129	0.963	0.8988	0.9269	0.8261
ACEP	0.9304	**0.9778**	0.9241	0.9367	0.861
SSGA-Model	0.9382	0.9724	0.9017	0.9747	0.8788
SSFGM-Model	**0.9389**	0.9732	0.9003	**0.9775**	**0.8804**

Note: The em dash symbol ("—") indicates that the data are not available.

limited standalone predictive power, provided valuable auxiliary structural context when integrated with other features. A comprehensive evaluation of these contributions is available in Supplementary Material Text S6.

Comparative analysis based on training and Benchmark Dataset 3

To ensure a more comprehensive evaluation, we extended our analysis to Benchmark Dataset 3, a larger and more diverse dataset. Detailed information on dataset construction and preprocessing is provided in Supplementary Material Text S3. We compared the performance of the SEQC_Model, SSGT-Model, SSGA-Model, and SSFGM-Model against SOTA models. The results demonstrated stepwise improvements (Fig. S1): (i) SEQC_Model outperformed other sequence-based methods; (ii) Structure-based models further enhanced classification accuracy, and (iii) The SSFGM-Model, which integrates structural and surface features, achieved the highest performance (ACC: 0.8866, AUC: 0.9486, MCC: 0.7749), underscoring the value of incorporating surface features. Detailed results are provided in Supplementary Material Text S5, Table S5. In addition, we further evaluated the robustness of the SSFGM-Model by testing its performance on AMPs with lengths exceeding the original training range (i.e., >50 amino acids). The results indicated that the model exhibited even better predictive performance for longer sequences. Detailed information can be found in Supplementary Material Text S8, Table S4, and Figure S4.

Overall, our framework outperformed other SOTA models regarding predictive ACC, AUC, and MCC. This superiority demonstrated the value of integrating sequence and structural information from multiple sources, such as residue contact maps and protein surface features, to enhance predictive performance. These findings reinforced the effectiveness of our approach and suggested that incorporating complementary features can lead to more comprehensive and accurate models for protein-related tasks.

Conclusions

In this study, we have comprehensively extracted multilevel characteristics of AMPs and compared model built on these features to elucidate the significant contributions of structural and surface geometric chemical properties in AMP identification. Our results demonstrate that deep learning models incorporating structural and surface features significantly outperform those based on traditional feature extraction methods. This work represents the first multimodal fusion of surface biochemical properties and geometric characteristics with an advanced protein language model at both the sequence and structural levels. Furthermore, we applied a geometric neural network alongside this fusion to predict AMPs, providing a novel methodology that integrates evolutionary, structural, geometric, and biochemical information of peptide sequences. Our findings underscore the efficacy of graph and geometric neural networks in AMP prediction, highlighting their potential for processing high-dimensional, complex biological data.

Nonetheless, several practical challenges remain. While ESM and ProteinBERT embeddings provide rich residue-specific features at the sequence level, they generate high-dimensional vectors that demand significant memory and computational resources. Additionally, structural predictions (via AlphaFold or ESMFold) and surface geometric physicochemical feature extraction (using MaSIF) substantially increase computational complexity and processing time. Moreover, although the current model excels at identifying AMPs, it lacks the granularity required to distinguish specific functional classes, limiting its direct applicability for detailed functional characterization. Despite these limitations, our integrated multilevel feature approach significantly enhances predictive accuracy and robustness.

Looking ahead, we plan to leverage this model to explore peptides with potentially unannotated/unlabeled antimicrobial properties, while extending its capability to differentiate between distinct AMP functional classes. Additionally, we also aim to integrate advanced generative models to design novel AMPs based on their sequence, structure, and surface features. Concurrently, we will optimize the feature extraction pipeline to improve computational efficiency and reduce memory requirements. These advancements will facilitate the development of peptides with targeted antibacterial activity and enhanced potency, accelerating the clinical translation of AMPs and inspiring innovative strategies for antimicrobial drug discovery.

Key Points

- We utilized AlphaFold2 to predict the structures of all peptides and subsequently extracted their surface geometric and chemical features, enabling a detailed characterization of each peptide's properties.

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- We developed a new approach by integrating graph neural network attention mechanisms with geometric neural networks to comprehensively analyze antimicrobial peptide characteristics. This represents the first application of surface geometric features to predict protein surface functional sites in the context of antimicrobial peptide identification.
- Benchmarking experiments demonstrated that our predictor, which incorporates both structural and surface features, outperforms current advanced models, achieving superior accuracy and robustness.
- This study provides a promising AI-driven framework for the discovery and design of novel antimicrobial peptides, paving the way for innovative therapeutic development.

Supplementary data

Supplementary data are available at *Briefings in Bioinformatics* online.

Conflict of interest: None declared.

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